ShinDengen

S3V80

General Rectifying Diodes
800V, 3.5A

Feature

- High Voltage
- Pb free terminal
- RoHS:Yes

OUTLINE

Package (House Name): AX14



Equivalent circuit



Absolute Maximum Ratings (unless otherwise specified : Tl=25°C)

Item	Symbol	Conditions	Ratings	Unit
Storage temperrature	T_{stg}		-55 to 150	°C
Junction temperature	T_j		-55 to 150	°C
Repetitive peak reverse voltage	V_{RRM}		800	V
Average forward current	$I_{F(AV)}$	50Hz sine wave, Resistance load, 20×20×1mm copper heatsink, On glass-epoxy substrate, Ta=40°C ※	3.5	A
Average forward current	$I_{F(AV)}$	50Hz sine wave, Resistance load, On glass-epoxy substrate, Without heatsink, Ta=40°C ※	2.6	A
Surge forward current	I_{FSM}	50Hz sine wave, Non-repetitive 1 cycle peak value, Tj=25°C	120	A
Surge forward current	I_{FSM1}	tp=1ms, sine wave, Non-repetitive, peak value, Tj=25°C	310	A

※ :See the original Specifications

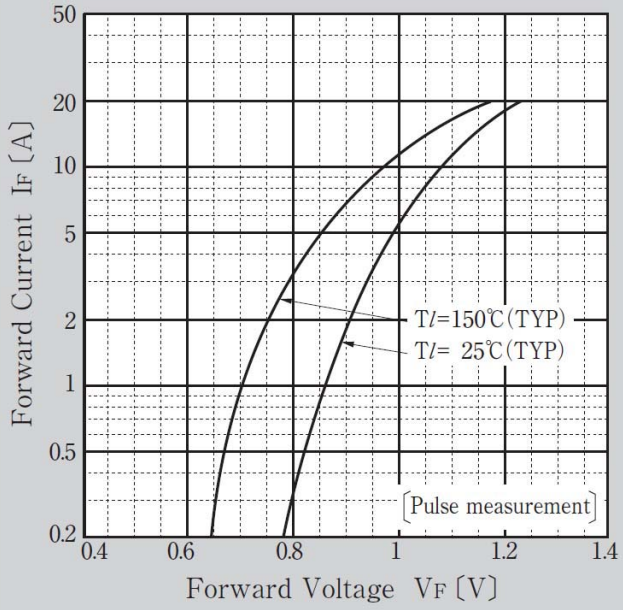
Electrical Characteristics (unless otherwise specified : Tl=25°C)

Item	Symbol	Conditions	Ratings			Unit
			MIN	TYP	MAX	
Forward voltage	V_F	IF＝2.6A, Pulse measurement			1.05	V
Reverse current	I_R	VR=800V, Pulse measurement			10	μA
Thermal resistance	$R_{th(j-l)}$	Junction to lead, On glass-epoxy substrate ※			6.5	°C/W

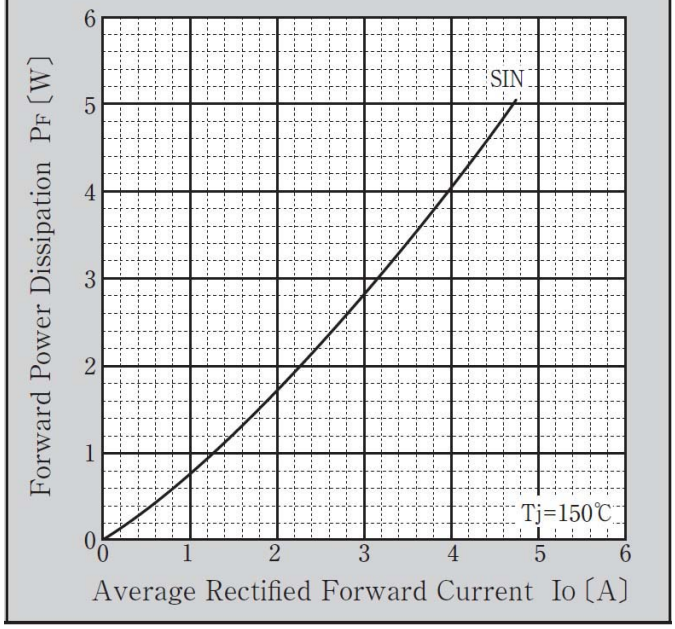
※ ：See the original Specifications

CHARACTERISTIC DIAGRAMS

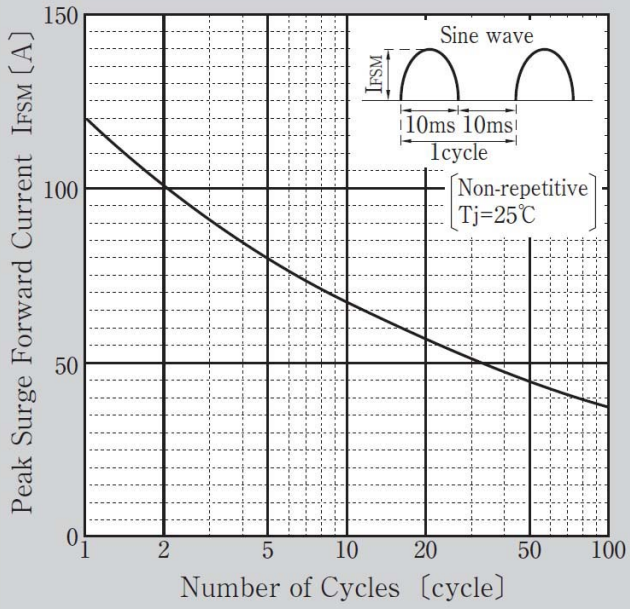
Forward Voltage



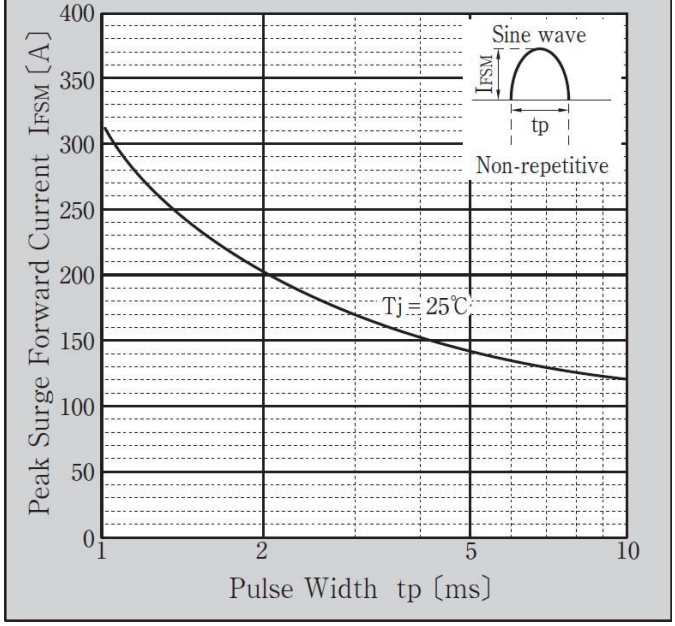
Forward Power Dissipation



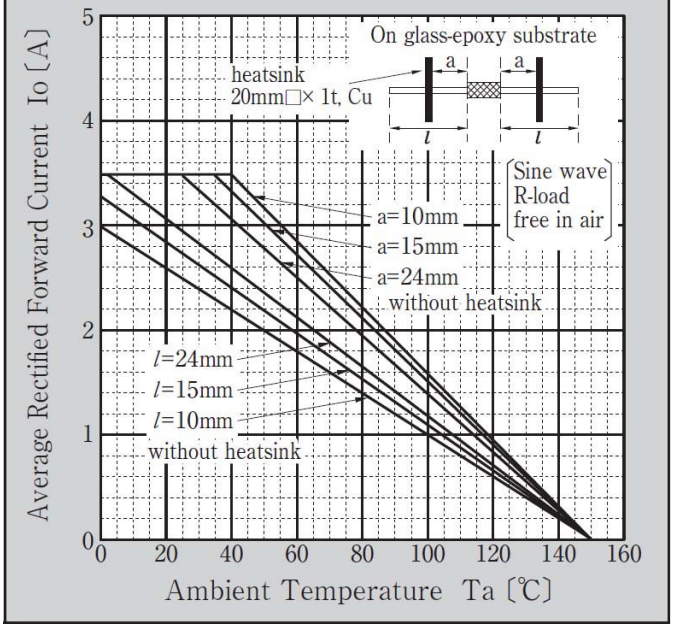
Peak Surge Forward Current Capability



Peak Surge Forward Current Capability



Derating Curve



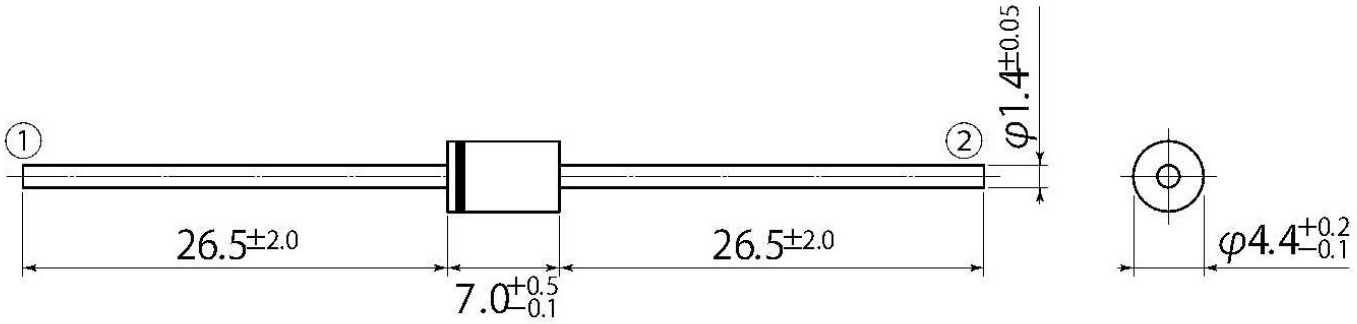
Outline Dimensions

unit:mm

scale: 2/1

A7

JEDEC Code	–
JEITA Code	–
House Name	AX14



Notes

1. If you wish to use any such product, please be sure to refer to the specifications issued by Shindengen.
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   【Special applications】
   Transportation equipment (vehicles, ships, etc.), trunk-line communication equipment, traffic signal control systems, anti-disaster/crime systems, safety equipment, medical equipment, etc.
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